

## MATINICUS ROCK, ME, USA

WMO: 994370

Lat: 43.7836N

Lon: 68.8549W

Elev: 53

StdP: 14.67

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 5.8        | 10.2 | -4.5                          | 4.3 | 9.6  | -1.5 | 5.1 | 12.8 | 51.8                  | 36.8 | 48.6 | 34.3 | 27.1                  | 320  | 0.895 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 6.3                    | 69.5            | 65.5 | 67.8 | 64.6 | 66.2 | 63.7 | 67.5                | 68.6 | 66.1 | 67.1 | 64.8 | 65.6 | 13.2                 | 220  |

| Dehumidification DP/MCDB and HR |       |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 67.2                            | 100.0 | 67.9 | 65.8 | 95.2 | 66.6 | 64.4 | 90.6 | 65.3 | 31.9          | 68.6 | 30.8 | 66.8 | 29.8 | 65.7 | 83.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| 48.0              | 41.7 | 36.5 | 1.4                        | 75.5 | 5.4                | 2.6 | -2.5                                               | 77.3 | -5.7       | 78.8 | -8.8       | 80.3 | -12.7      | 82.1 |  |  |
|                   |      |      | DB                         | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  |  |  |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 45.8   | 27.9  | 29.0  | 33.7  | 41.0  | 47.7  | 55.0  | 61.8  | 62.3  | 58.2  | 51.7  | 43.8  | 35.9 |
|                                                                               | DBStd       | 13.22  | 10.44 | 8.01  | 6.09  | 4.09  | 4.01  | 4.13  | 3.57  | 2.78  | 3.07  | 4.68  | 6.76  | 8.13 |
|                                                                               | HDD50       | 2830   | 686   | 587   | 506   | 271   | 95    | 8     | 0     | 0     | 0     | 37    | 201   | 439  |
|                                                                               | HDD65       | 7042   | 1151  | 1007  | 971   | 720   | 535   | 302   | 111   | 92    | 203   | 411   | 637   | 902  |
|                                                                               | CDD50       | 1283   | 0     | 0     | 1     | 0     | 25    | 156   | 364   | 382   | 247   | 91    | 15    | 2    |
|                                                                               | CDD65       | 19     | 0     | 0     | 0     | 0     | 0     | 0     | 10    | 9     | 0     | 0     | 0     | 0    |
|                                                                               | CDH74       | 7      | 0     | 0     | 0     | 0     | 0     | 0     | 5     | 2     | 0     | 0     | 0     | 0    |
|                                                                               | CDH80       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
| Wind                                                                          | WSAvg       | 18.4   | 23.2  | 21.8  | 21.3  | 19.0  | 17.2  | 15.0  | 14.0  | 13.8  | 13.3  | 18.3  | 21.0  | 22.3 |
| Precipitation                                                                 | PrecAvg     | 60.8   | 5.7   | 4.8   | 5.3   | 4.9   | 4.4   | 4.9   | 3.8   | 3.8   | 4.7   | 5.9   | 5.7   | 7.0  |
|                                                                               | PrecMax     | 82.2   | 10.5  | 10.3  | 8.9   | 9.8   | 8.7   | 10.0  | 7.6   | 9.2   | 9.9   | 16.5  | 9.8   | 11.6 |
|                                                                               | PrecMin     | 37.9   | 1.2   | 2.0   | 1.6   | 0.6   | 0.8   | 2.4   | 1.8   | 1.1   | 1.6   | 1.5   | 2.2   | 2.6  |
|                                                                               | PrecStd     | 8.6    | 2.1   | 1.6   | 1.9   | 2.0   | 1.9   | 2.0   | 1.5   | 2.0   | 2.1   | 2.9   | 1.7   | 2.2  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 50.4  | 45.4  | 48.5  | 54.0  | 63.5  | 69.9  | 73.4  | 71.5  | 67.4  | 63.7  | 58.0  | 56.7 |
|                                                                               |             | MCWB   | 50.1  | 44.5  | 45.5  | 46.1  | 57.7  | 63.9  | 66.6  | 66.3  | 61.9  | 58.2  | 55.3  | 55.2 |
|                                                                               | 2%          | DB     | 47.9  | 42.9  | 44.7  | 49.9  | 57.9  | 64.5  | 70.2  | 69.0  | 64.5  | 60.5  | 55.9  | 52.8 |
|                                                                               |             | MCWB   | 47.0  | 41.5  | 42.5  | 45.5  | 53.6  | 58.8  | 65.9  | 65.7  | 62.3  | 58.1  | 53.9  | 50.8 |
|                                                                               | 5%          | DB     | 45.3  | 41.4  | 42.4  | 47.7  | 54.9  | 62.0  | 68.1  | 67.6  | 63.4  | 58.7  | 54.3  | 50.2 |
|                                                                               |             | MCWB   | 43.2  | 39.9  | 40.2  | 44.1  | 51.8  | 58.1  | 64.9  | 65.0  | 61.5  | 56.9  | 52.6  | 48.7 |
|                                                                               | 10%         | DB     | 42.4  | 39.6  | 40.9  | 46.0  | 52.7  | 60.2  | 66.4  | 66.3  | 62.2  | 57.2  | 52.8  | 47.6 |
|                                                                               |             | MCWB   | 40.3  | 37.5  | 38.7  | 43.0  | 49.9  | 57.2  | 63.8  | 64.2  | 60.3  | 55.5  | 51.0  | 45.6 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 50.6  | 45.6  | 47.0  | 50.4  | 60.1  | 67.0  | 69.4  | 69.2  | 65.3  | 60.7  | 57.3  | 56.3 |
|                                                                               |             | MCDB   | 50.7  | 45.7  | 48.1  | 51.4  | 62.1  | 69.3  | 71.1  | 69.7  | 65.6  | 61.5  | 57.5  | 56.6 |
|                                                                               | 2%          | WB     | 48.1  | 42.4  | 43.6  | 47.3  | 55.3  | 61.3  | 67.7  | 67.5  | 63.9  | 59.3  | 54.8  | 52.3 |
|                                                                               |             | MCDB   | 48.3  | 42.7  | 44.8  | 48.5  | 56.9  | 62.8  | 69.5  | 68.1  | 64.1  | 60.2  | 55.5  | 53.1 |
|                                                                               | 5%          | WB     | 44.4  | 40.6  | 41.3  | 45.4  | 53.2  | 59.4  | 66.3  | 66.3  | 62.5  | 57.6  | 53.3  | 49.7 |
|                                                                               |             | MCDB   | 45.1  | 41.2  | 42.2  | 46.8  | 54.4  | 61.1  | 67.7  | 66.9  | 62.9  | 58.3  | 54.0  | 50.6 |
|                                                                               | 10%         | WB     | 41.4  | 38.3  | 39.5  | 43.9  | 51.0  | 57.9  | 65.0  | 65.0  | 61.4  | 56.3  | 51.6  | 46.5 |
|                                                                               |             | MCDB   | 42.7  | 39.5  | 40.5  | 45.3  | 52.6  | 59.5  | 66.1  | 65.7  | 62.0  | 57.1  | 52.7  | 47.7 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 11.1  | 9.9   | 7.9   | 5.9   | 6.3   | 6.9   | 7.1   | 6.3   | 5.2   | 5.3   | 7.3   | 9.9  |
|                                                                               |             | MCDBR  | 11.6  | 10.0  | 8.2   | 8.0   | 10.3  | 10.4  | 9.8   | 8.1   | 6.7   | 5.9   | 7.3   | 10.5 |
|                                                                               |             | MCWBR  | 12.9  | 11.0  | 8.1   | 5.9   | 6.9   | 7.1   | 6.9   | 6.0   | 5.6   | 6.0   | 8.6   | 12.0 |
|                                                                               | 5% WB       | MCDBR  | 11.0  | 10.4  | 8.2   | 7.1   | 9.9   | 10.0  | 9.0   | 6.9   | 6.2   | 5.7   | 6.8   | 10.3 |
|                                                                               |             | MCWBR  | 13.3  | 12.1  | 8.6   | 6.3   | 8.0   | 7.7   | 7.2   | 6.1   | 5.9   | 6.1   | 8.6   | 12.3 |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.309  | 0.313 | 0.332 | 0.356 | 0.384 | 0.423 | 0.444 | 0.413 | 0.369 | 0.342 | 0.323 | 0.304 |      |
|                                                                               | taud        | 2.476  | 2.483 | 2.472 | 2.434 | 2.400 | 2.304 | 2.266 | 2.343 | 2.443 | 2.514 | 2.532 | 2.522 |      |
|                                                                               | Ebn at Noon | 261    | 281   | 288   | 288   | 282   | 270   | 262   | 267   | 272   | 266   | 254   | 252   |      |
|                                                                               | Edh at Noon | 23     | 27    | 30    | 34    | 37    | 41    | 42    | 37    | 31    | 26    | 22    | 20    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 489    | 778   | 1166  | 1485  | 1689  | 1815  | 1906  | 1716  | 1382  | 895   | 548   | 396   |      |
|                                                                               | RadStd      | 37     | 62    | 79    | 154   | 196   | 155   | 133   | 85    | 82    | 82    | 50    | 45    |      |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (30 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +106  | N/A   |

Nomenclature: See separate page